SQL Explanation Grader: baseline vs RAG

Experiment **abstention_probe** · split **abstention_probe** (20 items) · generator **llama3.2:3b** · judge **qwen2.5:3b** (rubric) · generated 2026-10-26 18:07

How to read this report. "Judge approval" is the share of explanations an LLM judge graded good. It is an automated, unverified measurement, not ground truth. Deterministic checks are rule-based heuristics computed from the SQL structure. Agreement of the judge with reference labels is reported separately (results/rag/judge_agreement.json).

Headline comparison

Metric	Baseline (prompt only)	RAG
Judge approval (95% CI)	100.0% (20/20) [84-100]	100.0% (10/10) [72-100]
Approval, abstentions count as fail	100.0% (20/20) [84-100]	50.0% (10/20) [30-70]
Rubric pass (judge's own scores)	100.0% (20/20) [84-100]	100.0% (10/10) [72-100]
Mean correctness (1-5)	5.00	5.00
Mean completeness (1-5)	5.00	5.00
Mean clarity (1-5)	5.00	5.00
Mean groundedness (1-5)	5.00	5.00
Explanations with unsupported claims (judge)	0.0% (0/20) [0-16]	0.0% (0/10) [0-28]
Deterministic check coverage	0.99	0.97
Invented numbers (rule-based)	20.0% (4/20) [8-42]	10.0% (1/10) [2-40]
Raw SQL keywords in text	5.0% (1/20) [1-24]	70.0% (7/10) [40-89]
Abstentions	0.0% (0/20) [0-16]	50.0% (10/20) [30-70]
Generation errors	0	0
Mean end-to-end latency (s)	3.8	12.5
Mean retrieval time (s)	0.00	0.50
Mean input tokens	559	2743
Mean output tokens	95	273

Paired comparison (same items, both judged)

10 pairs. Both good: 10; both bad: 0; only baseline good: 0; only RAG good: 0. Approval difference RAG minus baseline: +0.0 points (95% bootstrap CI +0.0 to +0.0). Exact McNemar p = None.

Dimension	Mean RAG minus baseline	95% CI
correctness	+0.00	+0.00 to +0.00
completeness	+0.00	+0.00 to +0.00
clarity	+0.00	+0.00 to +0.00
groundedness	+0.00	+0.00 to +0.00

Grounding and retrieval (RAG condition)

Metric	Value
Answers with at least one citation	100.0% (10/10) [72-100]
Citation validity (cited source was provided)	1.000
Answers citing a source that was not provided	0.0% (0/10) [0-28]
Share of sentences carrying a citation	0.394
Lexical support of cited sentences (heuristic)	0.800
Model said evidence was insufficient	0

Metric	Value
Retrieval feature coverage	0.708
Evidence judged sufficient before generation	50.0% (10/20) [30-70]
Grounding status counts	{"abstained": 10, "grounded": 10}

Retrieval evaluation (no LLM involved)

Project-authored by Claude (AI-assisted) on 2026-10-26 by reading the section headings of each corpus document. Not human-reviewed yet. Treat retrieval scores computed with it as rule-based estimates, not human relevance judgements.

Chunk	Strategy	Mode	MRR	Recall@5	Prec@5	nDCG@5	Ctx recall	ms/query
400	decomposed	dense	0.824	0.925	0.556	0.616	0.892	161
400	decomposed	bm25	0.642	0.836	0.568	0.560	0.960	5
400	decomposed	hybrid	0.690	0.867	0.585	0.590	0.893	162
400	probe	hybrid	0.564	0.596	0.397	0.417	0.711	50
400	raw_sql	hybrid	0.691	0.797	0.522	0.538	0.799	58
200	decomposed	dense	0.784	0.899	0.571	0.584	0.943	161
200	decomposed	bm25	0.710	0.837	0.637	0.627	0.963	7
200	decomposed	hybrid	0.640	0.878	0.604	0.569	0.948	163
300	decomposed	dense	0.764	0.676	0.511	0.548	0.934	162
300	decomposed	bm25	0.658	0.842	0.597	0.576	0.962	5
300	decomposed	hybrid	0.644	0.834	0.535	0.531	0.940	160
500	decomposed	dense	0.694	0.727	0.448	0.500	0.873	160
500	decomposed	bm25	0.728	0.839	0.589	0.604	0.968	4
500	decomposed	hybrid	0.647	0.854	0.566	0.564	0.874	166
600	decomposed	dense	0.743	0.645	0.458	0.532	0.932	162
600	decomposed	bm25	0.705	0.835	0.548	0.571	0.913	6
600	decomposed	hybrid	0.615	0.830	0.490	0.502	0.899	160

Breakdown by SQL feature (dataset label)

Feature	n	Baseline approval	RAG approval	Base checks	RAG checks
CASE expression	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	1.00
CTE	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	1.00
EXISTS	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	1.00
GROUP BY with HAVING	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	0.75	1.00
JSON aggregate	1	100.0% (1/1) [21-100]	n/a	1.00	n/a
JSON function	3	100.0% (3/3) [44-100]	n/a	1.00	n/a
JSON table-valued function	1	100.0% (1/1) [21-100]	n/a	1.00	n/a
LEFT JOIN with IS NULL	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	1.00
ORDER BY and LIMIT	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	1.00
date function	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	0.75
math function	5	100.0% (5/5) [57-100]	n/a	1.00	n/a
scalar functions	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	1.00
subquery in WHERE	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	1.00
window function	1	100.0% (1/1) [21-100]	100.0% (1/1) [21-100]	1.00	1.00